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# An Attachment Model of Depression: Integrating Findings from the Mood Disorder Laboratory

Jeremy Holmes

This paper is written from a psychodynamic clinician's perspective, juxtaposing a psychoanalytic-attachment model of depression with recent developments in neuroscience. Three main components of the attachment approach are described: the role of loss, of childhood trauma predisposing to depression in later life, and failure of co-regulation of role of primitive emotions, such as fear, despair, and helplessness. Blatt's distinction between anaclitic and introjective depression is delineated and related to hyper- and de-activation of the attachment dynamic. Recent advances in evolutionary, sociological, epigenetic, biochemical, and neuro-imaging studies of depression are reviewed. A dynamic model of depression is proposed, linking interpersonal and intra-psychic perspectives with neuro-anatomical models. The final section of the paper considers the specific role of psychodynamic approaches to the treatment of refractory depression. These include length of treatment, capacity to rework implicit memories, and focus on transference and counter-transference.

After two decades in the wilderness, psychoanalytic therapies for depression are making a muted revival (Abass, 2006; Lemma, Target, & Fonagy, 2010), based in part on recent research in the basic science of mood (c.f. Luyten & Blatt, 2011). This paper examines attachment-informed psychoanalytic (Holmes, 2009) approaches to depression and its treatment in the light of current findings in the fields of genetics, neuro-imaging, neuro-endocrinology, and child development. It attempts to develop a model of mood disorder that does justice both to brain *and* mind, and to encompass the experiential phenomenology of depression. The aim is to

highlight the specific psychodynamic contribution to the understanding and treatment of depression and to claim its place as a relevant therapeutic player, alongside CBT and pharmacotherapy.

In the spirit of clinical relevance, I start with an example of depression as one might encounter it in the clinic.

Jane, 47, a married businesswoman with two children, was referred for psychoanalytic therapy by a physician because of medically unexplained abdominal pain. She had a near full house of Major Depressive Disorder symptoms, including weight loss, sleep disturbance, panic

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symptoms, low mood, poor concentration, and feelings of hopelessness. An obvious precipitant was the death of her mother some 8 months previously, although as the story unfolded it became clear that many of her symptoms predated this.

During the assessment phase of her therapy, she mentioned that she had also developed panic attacks at the age of 18 and, as though in passing, that this was after an “unwanted pregnancy.”

Therapist: “Unwanted by whom?” I asked.

Patient: “Oh, my boyfriend at the time was dead set against it.”

Th: “And you...?”

P: “Oh, I was longing for a baby, but he wasn’t having it.”

Th: “Indeed it was you, not he, who was ‘having’ it. I suppose what he ‘had’ was his way; you were left with the unmourned grief over a child that never was.”<sup><</sup>

Then came bitter tears, as though the loss had happened yesterday.

We shall return later to Jane, to the childhood relational constellations which predispose to depression in the face of loss, and the ways—including picking up on linguistic ambiguities—in which therapy can help overcome this example of “affect phobia” (McCulloch et al., 2003).

### A PSYCHODYNAMIC- ATTACHMENT MODEL OF DEPRESSION

In the section that follows, I shall present an attachment-derived theory of depression which builds on, revises, and extends aspects of the psychoanalytic viewpoint first put forward by Freud in *Mourning and Melancholia* (1917).

The attachment account of depression picks up on three principal themes.

## 1. The Centrality of Loss

True to his title, Freud starts from parallels between the phenomenology of mourning and depression. Depression follows actual or symbolic loss, both of which activate feelings of failure and lack of efficacy. But the reaction to loss in depression is both obscure and disproportionate:

[O]ne cannot see clearly what it is that has been lost... the patient cannot consciously perceive what he has lost... in the sense that he knows *whom* he has lost but not what he has lost *in him*... Melancholia is... related to an object-loss, which is withdrawn from consciousness, in contradistinction to mourning, in which there is nothing about the loss that is unconscious. (Freud, 1917, p. 246)

Here Freud moves from the outer to the inner world—from the world of psychiatry, one might say, to the world of psychodynamics. He goes beyond external loss to the *internal* sense of impoverishment characteristic of depression, confidently stating that in depression the loss is necessarily “withdrawn from consciousness.” But is this invariably so? Let’s go back to Jane.

Jane knew she had “lost” her mother. But what was it about that loss that felt so ir retrievable, made her feel so lost? At one level, she was fully aware of what was gone from her life. Her mother, who had lived next door, had been her constant companion, her confidante and best friend, the person she could turn to whenever things were difficult—in this case, when she developed abdominal pain. As a child, she recalled playing in the back yard and the panic feelings she experienced when she looked up in search of her mother’s face at the kitchen window—normally ever available for safety and reassurance and preoccupied with a chore—and she wasn’t there.

Her mother had been at the apex of her attachment hierarchy, and her husband,

even with all his good qualities, was definitely at number two or below. Thus, when on their honeymoon, Jane became temporarily deaf after a plane journey, her husband was dismayed that she had called her mother for reassurance rather than turn to him, as he had expected. In her depression, Jane's mother's absence left a vacuum of which she was painfully aware whenever she felt in need of succor.

A further twist emerged from Jane's history. Jane had been abruptly weaned at 3 months when her mother needed to attend to her own mother (Jane's grandmother) who became seriously ill at that time. One might postulate that Jane's vulnerability to separation and her fear of abandonment had their origins in this developmental trauma and its sequelae.

Jane's case is perhaps atypical in that depression and bereavement were in her case intertwined. But we can still ask of Freud's theory in what sense was Jane's loss hidden from consciousness? A moment in our work together illuminates this.

Jane felt that my probings about her relationship to her mother somehow carried the implication that she was making a great fuss, and that she should have "got over" her grief by now.

"I wonder how close *you* were to your mother," she declared. "Perhaps you just don't understand what it is like to have had such a loving and intimate relationship with one's Mum."

My slightly defensive reply went: "On the contrary, my feeling is that, although you are depressed and feel the physical pain in your stomach, you still haven't yet allowed yourself to experience (one might say haven't 'stomached') the full force of the mental suffering entailed in

losing the all-important person in your life."

At this, Jane burst into powerful sobs, going on to describe how for the first few months she had been unable even to look at her mother's house, and still now she could not bear to enter it to do the necessary sifting through and disposal of her possessions.

What was "hidden from consciousness" was not the *fact* of her mother's death, but its emotional import, the associated affect. This she kept literally (in the sense of not being able to enter her mother's house) at arm's length in her avoidance of exposure to any experience that might by-pass the fragile defenses she had built against feeling loss. Her transferential anger in response to my perceived heartlessness gave a glimpse of the inner cauldron of misery and rage that she was fighting so hard to keep at bay.

Without the support of her mother, Jane became helpless (Bibring, 1953) when faced with the threat that her stomach symptoms represented. An object relations and an attachment perspective would postulate that Jane's developmental history predisposed her to depression in that she lacked a robust sense of an *internal* "good mother/breast" (Klein, 1940) or "Internal Secure Base" (Holmes, 2001).<sup>1</sup> Her premature weaning experience may have been a significant first step in this adverse developmental process.

Jane's avoidance of anything associated with her mother graphically exemplifies "affect phobia"<sup>2</sup> (McCullough et al., 2003) as a central feature of depression. Helping to overcome affect phobia is a key therapeutic aim for a psychodynamic approach. Affects evoke phobic avoidance when the means to regulate them is lacking. Through the responsiveness of an attuned caregiver help-

1. The notion of the "internal mother" or "secure base" has to be seen as essentially metaphorical. There is no "mother homunculus" residing in the mind, but rather a self whose current state is shaped by past experiences of intimate relationships. Nevertheless the notion of an internal world populated by formative others remains a useful shorthand.

2. Affect phobia has the advantage of being a bridging concept between psychoanalysis and cognitive-behavioral conceptions of depression.

ing to co-regulate difficult feelings (sadness, hunger, rage, abandonment, etc.), securely attached children gradually build up the capacity for self-regulation (Schore, 2003). Initially such co-regulation is at a physiological level (cradling, soothing tone of voice, etc.), but with the acquisition of language, this becomes increasingly verbal (“you’re all right now,” “mamma’s here,” etc.). Self-soothing can then take off via internalized speech (saying to oneself, “this will be all right,” “we can cope with this,” etc.) (c.f. Meares, 2005).

In insecure attachment, this process is compromised. Free flow between affect, with its physiological components, and its verbal representation in the conscious mind is impeded. Affect phobia is clearly different from classical psychoanalytic repression, but nevertheless contains the idea that people avoid experiencing certain feelings (which are therefore “withdrawn from consciousness”), not necessarily because they are forbidden by a punitive superego (as Freud postulated), but because they are, in the absence of a secure base, too painful to bear on one’s own (Holmes, 2011).

Jane was in physical pain and felt “depressed,” but in depression there is often an *absence*<sup>3</sup> of feeling—no pleasure, no excitement, no hope, just a sense of deadness. For Jane to *experience* real sadness, for her somatized physical pain to transmute into psychic pain, she needed to feel safe enough in the therapeutic setting to do so, and that included trusting the analyst to tolerate her resentment at his lack of understanding. Jane’s paradox—which it was therapy’s job to help unravel—was that the one person who could have provided the safety to think about her loss was the mother whose death provoked the need to think about it in the first place.

## 2. The Role of Childhood Trauma

One of the most painful features of depression, both for the sufferers and those

close to them, is the way that the self turns against itself. This was the crucial issue that *Mourning and Melancholia* tried to unravel. In a famous passage, Freud argues:

Thus the shadow of the object fell upon the ego, so that the latter could henceforth be criticized by a special mental faculty like an object, like the forsaken object. In this way the loss of the object became transformed into a loss in the ego, and the conflict between the ego and the loved person transformed into a cleavage between the criticizing faculty of the ego and the ego as altered by the identification. (Freud, 1917, p. 248)

Here, Freud links childhood “loss” with later depression, and he accounts for the phenomenology of self-attack by suggesting that protest rightly belonging to the forsaking object is now transposed into the inner world. The attack on a part of the self is, via the “shadow” identification theory, an indirect attack on the abandoning object.

From a contemporary perspective, Freud’s theory is outdated for two related reasons: first, because it fails to differentiate between the normal developmental process in which all children have to cope with the loss of exclusive possession of the mother, as opposed to specific pre-depressive childhood trauma; and second, because Freud’s account ignores the actual behavior of a caregiver, seeing trauma and loss merely in terms of a putative narcissistic investment with which the child imbues the object, which in depression is withdrawn back into the self.

Nevertheless, implicit in the “shadow of the object” theory is the idea that childhood trauma can be the developmental precursor of adult depression. For Freud, the Oedipal situation is inherently “traumatic.” We now see childhood trauma as resulting from actual events—neglect, abandonment, or physical and/or sexual abuse. Brown and Harris (1978), Paykel and Herbert (1989), and more recently, for example, Artero and

3. Paralleling the absence of the loved one.

colleagues (2011) have established beyond doubt that loss plays a very significant role in depression. At least 70% of depressed women have experienced a significant loss event or chronic difficulties in the year preceding the onset of the depression, as compared with around 25% of non-depressed people. Nevertheless, it is possible to become depressed without obvious loss, and it is possible to experience major loss without becoming depressed. A probabilistic or “cascade” model (including the interplay between childhood trauma and genetic vulnerability, see below) is needed to account for illness in any particular case.

Brown and Harris (1978) found that vulnerability to loss depends on both current factors (lack of employment; having more than three children under 15; lack of a supportive relationship) and antecedent ones (early separation from or loss of mother). Parker (1979) showed that, as compared with controls, depressed women are more likely to view their mothers as having offered “low care and high overprotection.” Recent studies of depressed patients combining pharmacotherapy and psychotherapy (Zobel et al., 2011) suggest that early adversity is an important variable in predicting outcome, and that those with adversity (around 50% in the Zobel study) benefitted more from psychotherapy and antidepressants than if treated with pharmacotherapy alone.

Bowlby (1980) built his theory of depression on the Brown-Harris findings described above, emphasising the role of real loss and adversity, but also the helplessness (Bibring, 1953; Seligman, 1975) characteristic of the depressive state. He saw these in terms of an unresponsive environment in which the infant feels that whatever they do to try to elicit care makes little difference. This model encompasses the low self-esteem characteristic of depression in that self-esteem depends on feeling that we *can make a difference*, first to significant others in our environment, and then through a developing a sense of efficacy more generally.

For Bowlby, loss is experienced as irreversible separation. Separation from a secure base exposes the individual to risk; risk evokes fear. Anger at separation can be seen as a negative reinforcement schedule designed to reunite the separated one with the secure base—whether it is the baby crying for her mother or the parent reprimanding the child for risky behaviors. The lack of an answering response to these angry outbursts gradually persuades the bereaved person that the separation is irremediable and paves the way for despair, helplessness, and low self-esteem.

Attachment research (Dozier, Stovall-McClough, & Albus, 2008; McMahon, Barnett, Kowalenko, & Tennant, 2006; Murray, 2009) has confirmed the prevalence of non-secure attachment patterns in depression. Children with resistant insecure patterns tend to dampen down their separation protest when separated from their caregivers. The pain of separation is thus initially mitigated by its transposition from the external to the inner world. This protects the individual from the vagaries of fate, but a price has to be paid. Now the self is no longer nourished by its connection with loving others. The pain of the outer actual loss is replaced by guilt and inner emptiness, in which one part of the self attacks another.

In addition to emphasising the shift from an outer loss to a “loss in the ego,” Freud picked up on *conflict* between the self and its objects as a central theme in depression. The attachment account similarly sees insecure individuals in an ambivalence bind. They need proximity to the object for security; but if the object is inconsistent or abandoning, the attachment response will be one of protest and anger in an attempt to restore closeness. However, the insecure child fears that to express such protest runs the risk of further alienating the secure base, and at this point, depressive helplessness sets in.

People become depressed, this view suggests, because (a) they have been, or feel, abandoned, (b) they experience rage and outrage about this, and (c) they protect the ob-



ject from their rage by identification, taking the battle—in Jane’s case, with the mother who wasn’t there when she needed her—into the self, but (d) this anger toward the object arouses guilt, and depression then becomes a punishment befitting the crime. Finally, (e) if, in the “depressive position” (as opposed to the paranoid-schizoid world of depressive illness), one can “reinstate” the whole object in one’s inner world, in both its loving and abandoning aspects, then one is inured against further loss (Klein, 1940). If, through the transpositions and transferences of therapy, Jane could find, lose, and reinstate a good internal mother, forgivable despite having let her down through absence, she would be able to transcend her pain, physical and mental, and her locked-in inner battle with depression.

As therapy proceeded, Jane was gradually able to enter her mother’s house. A moment of symbolic “reinstatement of the lost object” occurred when she retrieved a favorite, but by now dusty dinner set used only at Thanksgiving, cleaned it, and displayed it with pride in her own kitchen display cabinet. Her other, lost in the outer world, was now firmly back in her mind’s eye.

### 3. Primal Emotions in Depression

Self-attack as diverted aggression cannot be taken as a general theory of depressive experience. A more nuanced picture, taking into account individual differences is needed. Blatt (2008) classifies depression psychoanalytically into “anaclitic” (object leaning) and “introjective” (self-absorbed) patterns. These map well onto attachment theory’s two forms of “organized insecurity”—hyper- and de-activating insecure attachment (Mikulincer & Shaver, 2008).

The de-activating or avoidant pattern renders its sufferers more prone to introjective ruminations. Such people will have had developmental experiences in which feelings were damped down (de-activated) in order to retain a degree of protection from a stressed

or rejecting caregiver (Ainsworth, Blehar, Waters, & Wall, 1978). As a way of surviving in a harsh world, these people may then imaginatively create an ideal inner scenario in which they are perfectly “good,” their caregiver perfectly attuned and responsive. When this fails, they then may turn against themselves in the way that the classical account describes. This sets up a typically depressive gulf between real self and ideal self; narrowing this discrepancy is a major therapeutic target (Kuyken et al., 2010).

In the dependent, “anaclitic” type of depression, however, self-attack and anger are usually far less prominent (Blatt, 2008). These sufferers often present a mixed depressive/anxiety picture. Jane’s intimate relationships illustrate Blatt’s anaclitic depression. Her attachment pattern was one of clinging adhesion to an inconstant or enigmatic object. This was played out in the transference, as she tried to maneuver the therapist into making life decisions for her, especially when it came to taking anti-depressants, the thought of which evoked in her an irrational fear.

“Please tell me whether I should take those pills or not,” she pleaded.

The therapist’s response was: “The psychiatrist in me thinks it would be a really good idea to give them a try; but as a psychotherapist I believe it’s vital for you to find the freedom to make decisions for yourself (and anyway, they’re unlikely to work unless you do!).”

Jane saw herself as helpless, rather than worthless, unable to interact effectively with her environment and make decisions for herself. Lacking a secure “internal mother” who could see her through moments of misery and doubt, she sought in her panic the physical presence of an external object (a perfect pill free from all side effects or an all-knowing doctor) in order to feel safe and able to offer absolute security in the face of her contradictory swings of emotion.

In sum, the attachment model postulates a loss event as the primary trigger for

depression. It then picks up on the procedural deficit associated with childhood insecure attachment patterns. The capacity to co-regulate the primal emotions associated with loss is compromised. The phenomenology of depression then reflects the overwhelming nature of these raw unmodulated emotions. Despair, panic, helplessness, and consequent sadism, rage, and revenge—primitively or primal emotions—hold sway in a psychic world unable to self-regulate.

In the remainder of this paper, I shall explore the extent to which contemporary neuroscience models of brain function in depression are consistent with this model, and in particular how the regulation, modulation, and mitigation of primal emotions can be disrupted. The overall aim is to juxtapose research findings with the attachment-informed psychodynamic model, in an attempt to build bridges between seemingly disparate realms of discourse. Integrative attempts of this sort run the risk of slippage between terms which seem to apply both to the interpersonal/experiential world of the psychodynamic clinician and the observational one of the neuroscientist. The connotations of words like “connection” or “circuit” differ depending on whether they refer to emotional links and feedback between individuals, their representation in the mind, or to neuronal transmission. Exploring the philosophical implications of this is beyond the scope of this paper (and competence of its author), but here the problem is at least acknowledged (c.f. Northoff, 2011).

## EVOLUTIONARY PSYCHOLOGY

Given the ubiquity and obvious fitness-reducing aspects of depression, a fundamental preliminary question is whether depression can be seen as adaptive, and therefore to have an evolutionary *function* (Neese,

2005). Three possible approaches are commonly described.

The psychoanalytic-attachment model proposed above is consistent with Simpson and Belsky’s (2008) “psychic pain hypothesis.” Sensitivity to physical pain is an essential protective mechanism for living organisms. Analogously, in the interpersonal environment, the capacity to experience psychic pain on separation ensures that no effort is spared to maintain the attachments essential for security and ultimate reproductive success. The capacity to become depressed in response to irretrievable loss is the dark side of the life-enhancing virtues of intimate attachment. Those that feel no psychic pain on loss may be protected from depression but risk deprivation through the superficiality of their attachments. Proneness to depression is thus the price paid for propensity to attachment.

A second popular idea derives from “rank theory” (Neese, 2005). Again the argument centers on the vulnerability associated with attachment.<sup>4</sup> As Tennyson (who suffered from serious depression when his beloved friend Arthur Hallam drowned at the age of 24) put it—“better to have loved and lost than never to have loved at all.” Attachments are essential; but they inevitably will sometimes be lost. However, in social groups where rank and dominance hierarchies prevail, loss lowers rank, rendering the sufferer vulnerable to reduced access to resources—food and reproductive opportunities. The symptoms of depression such as loss of appetite and reduced libido can be seen as adaptive in that by being temporarily *hors de combat* (out of action), the sufferer is no longer perceived by potential rivals as a threat, and he can be allowed to recover in peace.

Although depression may be an adaptive response to loss of status and rank, this view also suggests that societies where rank anxiety are prevalent will also have high

4. c.f. Francis Bacon’s famous description of a spouse and children as “hostages to fortune.”



depression rates. Recent studies showed a strong relationship between social disadvantage and the incidence of depression (Pickett & Wilkinson, 2010). Generally, the poor have higher levels of mental illness; this makes sense in terms of both the incidence of adverse life events (unemployment, housing difficulties, divorce and separation, death, illness rates) and the availability of mitigating support (high divorce/separation rates again, access to good medical services, including talk therapies). But the issue is not simply a matter of absolute poverty. Pickett and Wilkinson (2010) studied the impact of income differentials in advanced societies. They show that the greater the inequalities, the greater the levels of indices of social difficulty, including mental illness rates. They link this with the “affluenza” concept (James, 2007): inequality fosters “rank anxiety,” in which envy is rife; therefore, people are more prone to see themselves as “failures” or “losers” (a linguistic link with loss and depression), compared with more egalitarian and socially cohesive societies. Boehm (2009) and other evolutionary anthropologists see cooperativeness as an alliance of the weak against the strong, an antidote to dominance hierarchies, which, as the rank hypothesis suggests, can be a breeding ground for depression.

A third evolutionary view is the so-called “niche change hypothesis” (Watson & Andrews, 2002). Here, depression is seen as a necessary moratorium (Erikson, 1950)—an opportunity to review one’s situation, change direction if need be, and generally to *reculer pour mieux sauter* (retreat to jump better). Clearly, this is highly relevant to the work of psychotherapy.

While positively connoting depression in these evolutionary terms may be clinically helpful, and while depression in minor forms may be adaptive, the gross and severe forms that are psychiatry’s main business are perhaps better seen as extreme dys-regulations of these more adaptive depressive responses to loss. The purpose of therapy might then

be to help the patient get back on a productive rather than self-defeating depression track (Gilbert, 2006) or, as Freud (1893, p. 310) aphoristically put it, to transform neurotic “misery into common unhappiness.”

## SOCIAL PSYCHIATRY

We have already seen how some of the most important findings in mood disorders came from social psychiatry research (Brown & Harris, 1978). These have continued with the Newpin project (Lederer, 2009), an intervention study on the impact of befriending on depressed mothers in a deprived area of East London. Harris and her colleagues have shown that recovery from depression is fostered by the combination of befriending and a positive “new beginning” event, such as getting a job or starting a new relationship—an important reminder that life events can be antidotes to depression as well as precipitants. A neuroscience perspective (Panksepp & Watt, 2011) would see these new beginnings in terms of activating “seeking” and “playing” affective modules, as opposed to “rage” or “panic.” If therapy in Winnicott’s (1965) terms is “learning to play,” the replacement of negative primal emotions with more positive ones is a crucial therapeutic ingredient.

## EPIGENETICS

Epigenetics is strictly defined as the study of heritable changes in the phenotype transmitted by non-genomic means, but is often more widely referred to as the study of developmental pathways. The essential message is that current environmental adaptations can, under some circumstances, be directly transmitted to offspring. The exploration of epigenesis sheds new light on the nature-nurture debate, whose psychiatric

manifestation is the divide between “organic” and psychodynamic psychiatry.<sup>5</sup>

In the terms of this paper, the epigenetics of developmental psychobiology can be summarized by the equation:

$$\text{Phe} = \text{G} \times \text{E squared}^6$$

That is, a given phenotype (Phe) is determined by genetic makeup (G) “times” the environment (E) squared—“squared” because the environment impacts on the phenotype in two distinct ways. First, the Ec (the current environment), through promoter genes, influences the expression of the individual’s genetic potential. Second, the Ed (the developmental environment), through epigenetic mechanisms, inscribes itself in an individual’s biology, which is then transmitted to future generations.

Developmental neurobiologist Dr. Frances Champagne (2010, p. 564) puts it like this:

[T]here is emerging evidence that changes in gene expression both within the brain and in peripheral tissues are associated with *differences in the quality of the early environment* and that these developmental effects are maintained by epigenetic mechanisms that control the activity of genes involved in disease risk and behavioral variation. [my italics]

Epigenetics shows how, contrary to Darwinian dogma, acquired characteristics can be heritable. This is because the DNA base sequence is only part of the developmental story. DNA is tightly bound in the cell to histone proteins. DNA is only available for transcription by mRNA into protein if the binding is sufficiently loose. It is now known that environmental influences impact directly on histone binding in certain tissues, especially the psychobiologically relevant hypothalamus. Thus, whether or not a par-

ticular gene is expressed depends on environmental influence, and this is then transmitted through the mother to future generations.

In addition, different environments switch on or off promoter genes, thereby determining varying phenotypic manifestations. Champagne’s experimental model is maternal grooming behavior in rats (Champagne, 2010; for reviews, see also Goldberg, 2009; Polan & Hofer, 2008). In females, maternal care in one generation directly influences the ability of their pups to care for their own infants in the next. “Good” rat mothers show a high degree of “LG” (licking and grooming) behavior with their pups, who in turn typically show less fear when presented with novelty as adults, when compared with offspring of low LG mothers.

The subtle interplay between gene expression and current and developmental environment is illustrated by a study in which rats were put into one of three rearing environments: “enriched,” “impoverished,” or neutral (Champagne, 2010). Those in the impoverished environment did not develop high LG behaviors even if they had a high LG mother; those in the enriched environment developed LG behaviors even if their mothers had been low LG.

Thus, the “distal” environmental conditions over-rode “proximal” maternal behaviors. Only in the neutral environment did LG “breed true,” so that low LG mothers had low LG offspring and high LG mothers high LG offspring. But it was in those environments that high LG mothers, when rearing offspring of low LG mothers, could transmit high LG to the children and grandchildren of those fostered pups, presumably by epigenetic mechanisms. Significantly (see below), the low LG offspring showed accelerated sexual maturation compared with the high LG-reared group. This suggests that the environment plays a crucial role in enhancing or suppressing genetic potential, and that

5. Bowlby always maintained that attachment theory, rooted as it is in evolutionary biology, was no less “organic” than psychopharmacology.

6. This tongue-in-cheek formula is based, of course, on the rhyme-scheme of  $e = mc$  squared.

this operates both distally (impoverished or enriched) and proximally (maternal licking behaviors).

The biochemical basis for these effects involves DNA methylation, which, by affecting gene-protein transcription rates, can alter the glucocorticoid receptor responsiveness in the hippocampus. Stress responses are modulated through the HPA system. Stress induces reductions in hippocampal volume. This in turn means that the feedback loop between the adrenal cortex in the pituitary is disrupted, leading to chronically elevated cortisol levels. Rodent data suggest that perinatal environmental conditions can result in long-term silencing of promoter genes through DNA methylation, and that this epigenetic modification results in changes to HPA responsiveness persisting into adulthood (Weaver et al., 2005). In sum, when a stressed mother's HPA system is compromised, her offspring have similarly abnormal stress responses.

Similar processes operate in humans. Exposure to abuse in infancy is associated with altered levels of DNA methylation in hippocampal tissue (McGowan et al., 2009). Oberlander and colleagues (2008) found elevated neonatal GR promoter methylation levels in infants born to depressed mothers. The methylation of the neonatal GR promoter predicted increased salivary cortisol levels of infants—a measure of “stress”—at 3 months.

Epigenetic variability in response to stress, in both humans and rodents, can be conceptualized as adaptive. The harsher the environment, the more short-term survival strategies take precedence over the comparative “luxuries” of attentive nurture and exploration. In adverse environments, high reactivity (adaptive misperception: “take any unexpected sensation as potentially threatening”) and early sexual maturation (when life expectancy is reduced) become priorities. Propensity to depression associated with insensitive caregiving is therefore an unavoidable side effect of these survival strategies.

This analysis applies equally to the seminal work of Caspi and colleagues (2003) on the role of the serotonin transporter gene polymorphism (5-HTTLPR) in depression. Possessing one or both copies of the s (“short”) allele predisposes to depression in relation to stressful intercurrent events, compared with those with the l (“long”) allele. The combination of childhood maltreatment/low care and possession of one or both short alleles is a quantitative predictor of depression in adult life (Artero et al., 2011). Despite initial difficulty in replication, recent meta-analysis has amply confirmed the original findings of Caspi and colleagues (Karg, Burmeister, Shedden, & Srijan, 2011).

Thus, the immediate reactivity that poor care elicits may be adaptive in that such infants need to be able to latch onto whatever care is available, however suboptimal, and be self-protectively hyper-reactive to adverse experiences, even if this stores up future trouble. The recent finding that insecurely attached girls at one year have earlier menarche compared with their secure counterparts underscores this point (Belsky, Houts, & Fearon, 2010). Insecure attachment instantiates psychobiological processes, presumably by epigenetic means that will not manifest themselves until the onset of menarche.

Steele and Seifer (2010) have studied another epigenetic locus, DRD4 polymorphism, known to represent a risk for attachment disorganization (in which depression is also often a feature). While being a risk factor, DRD4 is also a marker of high responsiveness to therapeutic interventions designed for toddlers with aggressive behavior problems. It seems that the same genetic polymorphism that represents risk also confers protection.

The DRD4 allele may indicate an increased *biological sensitivity to context* with potential for negative health effects under conditions of adversity (e.g., unresolved mourning regarding past loss or trauma in the mother) and positive effects under conditions of support and

protection. (Steele & Siefer, 2010, p. 63; author's italics)

Discussing both the 5-HTTLPR and the DRD4 findings, Belsky and colleagues (2009) suggest that “vulnerability” genes be re-conceptualized as “plasticity” genes, rendering the individual more susceptible to harm in adverse environments, but also better able to profit from favorable ones.

The relevant conclusions for the psychodynamic clinician from these epigenetic and genetic studies can be summarized as follows:

- a. We need to think in terms of cascading *developmental trajectories* in which the interplay of genes, the social environment, both proximate and distal, and the part played by mitigation, reversibility, and compensatory factors are all taken into account.
- b. Extrapolation from animal data suggests that suboptimal parenting in infancy may increase depression vulnerability in the next generation via decreased exploratory behavior and heightened reactivity. These features may be transmitted epigenetically to the third generation and beyond. Thus, therapeutic narratives need to take the grandparental generation into account.
- c. There may be a paradoxical but benign relationship between vulnerability/reactivity and responsiveness to psychotherapy. Biologically sensitive individuals, whether through classical genetic polymorphisms or epigenetic transmission, are more vulnerable to environmental trauma, but also to psychotherapeutic amelioration.
- d. Given neuroplasticity and the fact that the environment can impact directly

on biological systems, childhood adversity is potentially amenable to psychotherapeutic intervention, but this may need to be at a relational and prolonged level if permanent change to HPA axis reactivity is to be achieved.

#### NEUROIMAGING: THE BRAIN IN ACTION

Neuropsych psychoanalysis (Northoff, 2011; Solms & Turnbull, 2002) is an emerging discipline in our field. Aware of the limitations of a purely computational approach, neurologists have been drawn to psychodynamic models capable of linking experiential meaning with structural models of the mind.<sup>7</sup>

Neuropsychiatrist Helen Mayberg (2009) sees pathology in depression as entailing two related areas of dysfunction: (a) impedance to the normal integrated flow of cerebral activity across regions, and (b) failure of the regulatory function of one part of the brain over another.

The structural areas implicated in depression include the dorsolateral pre-frontal cortex (DLPFC), cingulate cortex, basal ganglia, amygdala, and hippocampus (Mayberg, 2009). Three sets of findings stand out as particularly significant from the perspective of the psychodynamic clinician:

First is the well-replicated consensus that there is *decreased dorsolateral pre-frontal cortex (DLPFC) activity* in depression (Wang et al., 2008), as measured by blood-flow and glucose metabolism. This relationship is quantitative, with inverse relationship between severity of depression and prefrontal cortex function (Brody et al., 2001). The principal role of the DLPFC lies in planning and regulation of emotions.

Second, ventral prefrontal cortex (VPFC) and amygdala *hyperactivity* has

7. Psychoanalysts, in turn, searching for scientific credibility, have been beguiled by the vivid imagery and circuitry of their neuroscience colleagues.

been consistently reported, especially where depression is associated with anxiety and obsessionality (Osuch et al., 2000). The amygdala, fully developed at birth and functionally connected to the hippocampus, is at the center of the fear regulation system and is particularly involved in implicit as opposed to episodic learning of intense, emotionally charged experiences (LeDoux, 1994). The significance of this will be developed below.

A third region relevant to the genesis and treatment of depression is the subcallosal cingulate gyrus, and in particular area Cg25. Severe depression is associated with excessive neural activity in this region, as are sad mood states in normal subjects (Mayberg, 2009). Responders to SSRI antidepressants show reductions in blood flow in Cg25 as their depressive symptoms remit (Mayberg et al., 2000).

A speculative model combining these findings with the attachment picture proposed in the first part of this paper would suggest the following:

1. Decreased DLPFC activity is consistent with the clinical observation of depressive disengagement from the world, moving from an object relations to a self-sufficient, “narcissistic” mode of functioning. Carhart-Harris, Mayberg, Malizia, and Nutt (2008) describe a “default mode network” (DMN) pattern of brain functioning that is superseded when the subject is engaged with some external task. In depression, the DMN takes over, and this can be conceptualized as the replacement of self-other relating (and associated neural circuitry) with endogenous self-self circuits.
2. Hyperactivity in the basal ganglia and amygdala can be seen as reflecting “damming up” of activity arising from limbic system centers. With the DLPFC in DMN mode and diminution of planning and regulation, there is a failure to subject primal affects to regulation and reality testing. With the brain in DMN mode, feelings cannot be brought into play in the interpersonal field—hence, the salience of “primitive emotions” in depression.
3. Cg25 can be postulated as a repressive agency, attempting to damp down and keep at bay impulses from lower centers (Carhart-Harris et al., 2008), which threaten to overwhelm the under-functioning higher centers, such as the DLPFC (c.f. Mayberg, 2009). The Cg25 center acts in a superego-like fashion, punitively self-persecuting as Freud postulated, vainly attempting to maintain a degree of psychic equilibrium.
4. Given that the amygdala matures prior to the hippocampus, and given amygdala hyperactivity in depression, implicit (as opposed to episodic) memories of childhood trauma may be revived in depressive states (Renn, 2011). The salience of these ingrained procedural memories may make sufferers with a history of childhood trauma less amenable to short-term therapies (Zobel et al., 2011). The latter target cognitive processes, which are de-activated.

This model predicts that recovery from depression will entail a normalization of the DLPFC, Cg25, VPFC, and amygdala dynamic balance. Depressed people who respond to antidepressants do indeed show concomitant increases in DLPFC activity and decreases in Cg25 blood flow (Mayberg et al., 2000). However, the converse is found in CBT responders, who show decreased DLPFC blood flow as they get better. Viewing brain function as a dynamic system makes sense of this apparently paradoxical effect. CBT encourages patients to use their rational cognitive capacities to re-evaluate the primitive emotions that dominate in depression (Kuyken et al., 2010). It is likely, therefore, that they will move from high to low PFC



blood flow as equilibrium is restored, after an initial burst of PFC activity. This would be particularly true for Blatt's "introjective" group, where successful treatment will need to reduce the tendency to rumination and may entail long-term therapy (Leischenring & Rabung, 2011).

### AN INTEGRATIVE MODEL FOR DEPRESSION

I shall now put forward a psychoanalytic-attachment model of depression incorporating the above findings and offering some possible mechanisms of action for psychoanalytic psychotherapies.

1. In a normal resting state, the individual and her/his attachment figures are in a dynamic equilibrium, in which the security and responsiveness of the environment facilitates exploratory physical and mental functioning (including exploring one's own and others' mental states, that is, "mentalizing"; Lemma et al., 2010). The neural substrate for this entails interacting intersecting feedback loops between the DLPFC and a) the interpersonal environment, that is, "significant others" and *their* DLPFC, and b) the intrapsychic environment, represented by the "lower" centers, including the amygdala. Primal emotions, evoked by external events including loss, are filtered through the "gateway" of the Cg25 and processed—regulated, moderated, and mentalized—by the DLPFC.
2. The establishment of a "good internal object" is the result of repeated cycles of need, loss, and repair throughout childhood. This leads to the instantiation of an internalized emotion-processing attachment representation, based on "good enough" security provision by the external environment in the course of emotional development.
3. The system of intersecting feedback is disrupted by significant loss events, whether actual (e.g., loss of an attachment figure through death or divorce) or symbolic (e.g., loss of self-esteem following a career setback). This disrupts the normal attachment response to loss, in which the primitive emotions of mental pain trigger seeking out comfort from a secure base. But if the sought-after comforter (actual or symbolic, e.g., career advancement) is the very source of security that has been lost, this leads to a turning away from the external world.
4. In the normal grieving process, there is still sufficient "ego strength" for the sufferer to replace the lost external relationship with an internal "good object," enabling the primitive emotions of mental pain, anger, guilt, and so forth generated in the lower centers to be processed. Adaptive repression plays a part too: under the aegis of Cg25, pain is modulated and dealt with in manageable quanta. In addition, the supportiveness of the environment and availability of potential new attachment figures during and after grieving significantly influence outcome.
5. Depressive illness, rather than normal grieving, may result following loss, depending on developmental factors: (i) an insecure rather than a secure attachment relational disposition which compromises the capacity of the DLPFC to regulate primitive emotions, and (ii) a history of childhood trauma or neglect, either in the individual or, epigenetically, in the previous maternal generations. This "sets" the HPA axis in ways that lead to excessive reactivity, with the potential for triggering early trauma schemata by current loss events.



6. Dysregulation after loss will follow different patterns depending on the individual's attachment style. In the deactivating (avoidant) type, the subject withdraws from the environment and tries to deal with primitive emotions by repression. Here, excessive activity in Cg25 leads to an introjective pattern, with typical self-reproaches and rumination as elaborated on by Freud. In the contrasting hyperactivating (ambivalent) pattern, the depressed person becomes anaclitic, seeking reassurance, becoming overly dependent, and attempting to use the clung-to other as an auxiliary ego in an effort to process the primitive emotions with which she is flooded.
7. Therapy helps move the depressed individual back onto a less dysregulated pathway and helps to restore normal responses to loss and setback in a number of ways:
  - (a) A stable, secure collaborative object relationship, which temporarily takes over some of the functions of the lost object, is offered.
  - (b) The cognitive processing of emotions is fostered, thereby overcoming affect phobia. In the introjective pattern, the intrapsychic dynamic is opened out, lessening the repressive grip of the superego/Cg25, allowing feelings to be felt and processed, with the auxiliary help of the therapist.
  - (c) In the anaclitic pattern, the therapist gradually returns projected feelings back to the sufferer, re-empowering her DLPFC's autonomous capacity to regulate her/his emotions.
  - (d) The intensity of primitive emotions "from below" is reduced. This can happen in a number of ways, including the gradual processing of implicit memories of trauma in psychodynamic therapy; antidepressant therapy; and meditative techniques in MBCBT (Kuyken et al., 2010). In each case, there

may be a gradual resetting of the HPA axis, with reduced glucocorticoid sensitivity, and thereby enhanced skillfulness in using cognition to modulate emotion.

(e) Finally, there is the instantiation of hope, the possibility of a new beginning, and the reduction of helplessness (Bibring, 1953). The lost object is restored to the inner world, which is thereby strengthened in parallel with a renewed interest in external reality. Hope can be equated with the relinquishment of self-absorption (and DMN function), and with a renewed trust in fruitful exploration, bolstered by the knowledge that security will be available if loss or threat once more supervenes.

### PRACTICE POINTS

In this survey, I have deliberately avoided a focus on outcome research and the relative merits of different therapeutic approaches.<sup>8</sup> Nevertheless, the model proposed above has significant implications for psychoanalytic practitioners.

First, depression needs to be seen as a heterogeneous condition (Blatt, 2008), both in terms of severity and subtypes of depression. A stepped care model (NICE, 2005) seems appropriate, in which CBT's combination of a good therapeutic relationship and reactivation of cognitive function in the DLPFC may in many cases suffice to restore psychological equilibrium. Where there is a history of early trauma or neglect, the prognosis tends to be less good and the indications for a psychodynamic approach correspondingly greater (Leischenring & Raybung, 2011; Zobel et al., 2011). It is important to bear this diagnostic heterogeneity in mind within the psychoanalytic paradigm, in that the standard psychoanalytic model of depression of "anger turned against the

8. But note, among others, Shedler (2010) and Abass (2006) for robust evidence for psychoanalytic treatments for depression.

self” is more applicable to introjective than to anaclitic patterns.

Psychodynamic practitioners will consider what the unique contribution of their discipline might be to the treatment of depression. Three points stand out. The first, while not specifically psychoanalytic in itself, is length of treatment. There is accumulating evidence for the superiority of outcomes for longer- over short-term therapies, although differences do not show themselves until well into the third year of treatment (Knekt et al., 2008). Despite evidence for neuroplasticity, in severe cases prolonged treatments will be needed if the biological substrate for depression is to be reset in less pathogenic ways. Lessening the strength of primitive emotions, reducing the need for excessive repression, decreasing the reactivity of the HPA axis, and fostering the capacity of the mind to regulate and process painful feelings, all take time.

Second, psychoanalytic approaches can help transmute and modify implicit memory systems that are likely to be awry in individuals who have experienced trauma or neglect in early childhood. Within the boundaries of the setting, therapy offers new experiences of sensitivity, responsiveness, and the capacity for reflection in place of confusing, disorganizing, and controlling interpersonal patterns of relating. Becoming aware of these ingrained patterns, as well as providing a setting in which they can be reworked, is central to the mechanism of action of psychoanalytic therapy (c.f. Fonagy, 1998; Holmes, 2010, 2011; Renn, 2011).

This leads, finally, to the unique and specific contribution of psychoanalysis to psychotherapy practice—working with the subtleties of transference and counter-transference. Let us return to Jane and the vignette with which we started.

The presence of affect phobia was signaled in Jane by a verbal ambiguity. The double meaning of “he wasn’t having it” exemplified how repressed primal emotions make their presence felt and the therapist’s role in identifying these

markers of repressed meaning. The implicit message from the therapist and the therapeutic situation was: “Your painful feelings are present; you aren’t consciously aware of them, but they manifest themselves in your speech patterns and perhaps in your somatic symptoms. It is now safe to feel those feelings in the secure presence of a therapist.”

Jane was in a dilemma about taking antidepressants. The vacuum left by her mother’s death meant that she found it difficult to make decisions for herself, and she tried to push the analyst into deciding for her, which he initially resisted. At the end of one session, stung by her indecisiveness, and probably in a minor enactment, he found himself encouraging her to stop vacillating and to give them a try.

Jane returned the following week, admitting that she still hadn’t been able to bring herself to start them. Her demeanor seemed shamefaced and guilty. When the analyst pointed this out, she replied, “Oh, I thought you’d be really cross with me.” While noting his counter-transference of mild exasperation, the analyst then explored a possible parallel between the sense that Jane felt she had to please him, that he would be annoyed if she didn’t take her pills, and the way she and her mother had related. Now, for the first time, ambivalent feelings about her mother surfaced. Yes, mother was her best friend, *but* she could be demanding and difficult at times; if Jane crossed her in any way, she would sulk. Then Jane would feel utterly wretched and abandoned, and compelled to break the ice and say she was sorry, even if she knew in her heart that it was her mother who was in the wrong.

The minutiae of transference led to a formulation as follows. Jane’s relationship with the therapist revealed a picture of anaclitic depression arising out of a hyper-activating pattern of insecure attachment. Jane’s maternal caregiver was inconsistent: sometimes she was available or even overly available, sometimes mysteriously absent and rejecting. This set up a pattern of clinging attachment in which Jane’s self-composure was dependent on proximity to her object. When

her mother died, this was not just a present loss, but it revived earlier feelings of abandonment when her mother had withdrawn her love when Jane stepped out of line. There may also have been procedural residues of premature weaning and her mother's anxieties about the loss of her own mother.

Jane's underlying resentment and concomitant fear of "biting the hand that feeds" underlay her defensive idealization of both her mother and the analyst. He was the "one supposed to know" (whether or not to take antidepressants), and yet she felt he really knew nothing about her (her fear of side effects, her wish to solve her problems without dependency, etc.). All this transmitted itself counter-transferentially to the analyst in the minor enactment in which he exasperatedly offered advice rather than interpretation. But by bringing Jane's resentment into the open, it was possible for the underlying feelings of fear, panic, helplessness, and rage to be processed by two DLPFCs acting in concert.

In order for Jane to benefit from therapy (antidepressants aside), she would need to co-create a pattern of relationship in which the analyst was available but not controlling, and to be able to express disagreement and resentment towards him without feeling that it would lead to rejection. Strengthened in this way, she would be more resilient to the stresses and losses that the future might bring.

Implicit in the above is the idea that the neurobiology of transference in depression depends on hypofrontality of the DLPFC. Instead of the normal interlocking of two mutually exploring minds, the sufferer's appraisal functions—of both self and other—are inhibited, so that primal emotions, including implicit procedural memories, biologically inscribed via epigenesis, are either projected onto or into the therapist (anaclitic pattern) or sequestered within the pressure cooker of the psyche (introjective pattern). Withdrawn into herself, Jane could only see the analyst in terms of pre-existing mental structures. In both cases, the therapeutic task is to re-activate the DLPFC and thereby re-

duce affect phobia. In the anaclitic pattern, difficult feelings are gradually returned from therapist to the sufferer for reappraisal; in the introjective pattern, the patient is encouraged to draw on the therapist's support in order to mentalize and modulate the hitherto warded-off mental pain and rage.

An example of Jane's progress was given in the "reinstatement" of her mother's dinner set. Another occurred when she applied for a job for which she was not really qualified and, unsurprisingly, failed to get it. Jane was immediately plunged into depression, self-doubt, and feelings of inadequacy. However, that night she dreamed that she was climbing a ladder to change a light bulb, with her mother at the bottom of the steps telling her to come down, that she was about to fall. Discussing this dream with her analyst, she saw how undermining her mother's "over-helpfulness" had been and how it inhibited her attempts to change. Processing her feelings of depression in this way (using her own and the analyst's DLPFC to regulate her subcortical affect) seemed to alleviate her depression and enabled her to laugh at her thus far ill-thought-out attempt to find a job.

## CONCLUSIONS

In moving from an external loss to its meaning and significance in the inner world, Freud was moving from psychiatry to psychoanalysis, from explanation to meaning. It could be argued that in exploring the biological and scientific, as opposed to the existential dimensions of depression as we have done here, this process is put into reverse and an essential aspect of the psychoanalytic project has been lost. As Lear (2003, p. 146) puts it:

We can learn much about the brain that is of value in alleviating human suffering... but none of this can answer the subjective question: what is it *for me* to become a person? Psychoanalysis is a process by which I come to take respon-

sibility for hitherto unconscious aspects of myself. I thereby deepen myself as a subject.

The thrust of this paper has been to suggest, to the contrary, the more we under-

stand ourselves as biological beings, and the more conscious and objective we are about both brain *and* mind, and how they dovetail one with another, the deeper and more responsible our subjectivity can become.

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